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# A continuum damage model for three-dimensional woven composites and finite element implementation

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ABSTRACT

A continuum damage model for predicting the damage initiation and development in three-dimensional (3D) woven composites is proposed, in which the fiber fracture, inter-fiber fracture and matrix fracture are considered in the level of the fiber yarn and the matrix. A set of damage variables are presented to characterize all the failure modes of the fiber yarn and matrix. The damage initiation and propagation criteria are based on the Puck criteria for the fiber yarn and the paraboloidal yield criterion for the matrix. This continuum damage model is implemented by combining it with the finite element method (FEM). To validate the continuum damage model, the quasi-static tensile experiments of a type of 3D woven composites are carried out, and the tensile failure is predicted by the damage model which agrees well with the experimental results.



## 1. Introduction

3D woven composites are typical fiber reinforced material which were first developed nearly 40 years ago [1]. Compared with the composite laminates, 3D woven composites can produce complex near-net-shape preforms and have higher delamination resistance and damage resistance.

The failure criteria are the basis of the failure prediction of 3D woven composites. Tsai-Hill criterion [2] was proposed for anisotropic material first and it is the promotion of the von Mises criterion. After that Tsai-Wu criterion [3] and Hoffman criterion [4] were put forward. All of these were classical theories and not able to distinguish the failure modes. Differently, Hashin criterion [5] is formulated under tensile and compressive loads in longitudinal and transverse direction of fiber, respectively. Hashin criterion was improved by Sun et al. [6] and Puck et al. [7,8]. The inter-fiber fiber fracture and inter-fiber fracture are discussed detailedly in the Puck criteria. Davila et al. [9] proposed a failure criteria named by LaRC03 for composites laminates which can predict matrix and fiber failure accurately. Catalanotti et al. [10] proposed a fully 3D failure criteria for polymer composite reinforces which considered the effect of ply thickness.

The continuous damage mechanics (CDM) approach is an important tool for modeling damage evolution in the fiber reinforced composites. Gorbatikh et al. [11] presented a new property degradation procedure to model the damage evolution in textile composites. Bahei-El-Din et al. [12] proposed a damage progression model for multi-scale analysis of 3D woven composites. The RVE of the woven material derived from micrographs is used in his model. Greve and Pickett [13] presented a numerical material model for intra-laminar failure prediction of biaxial non-crimp fabric composites considering effects of fabric preshear. Maimí et al. [14,15] proposed a continuum damage model for composite laminates. Fang et al. [16] investigated the compressive properties of the 3D braided composites using the damage theory. Chen et al. [17] proposed a combined elasto-plastic damage model for progressive failure of the fiber-reinforced composites. Bogdanovich et al. [18] presented an experimental study on the damage in the 3D orthogonal woven composites. Cousigné et al. [19] developed a nonlinear numerical material model for textile composite materials considering post-failure damage. Melro et al. [20] developed an elasto-plastic thermodynamically consistent damage model for the matrix in the fiber reinforced composites. Martín-Santos [21] developed a continuum constitutive model for the simulation of fabric-reinforced composites based on CDM.

Most of the above damage models are proposed for 2D composite laminates. As for the limited simulation investigations of the damage evolution behavior in the 3D fiber reinforced composites, some typical limitation can be found in the previously published papers. For example, (1) most of the adopted classical criteria were not able to judge the initial damage precisely and effectively; (2)

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the damage and failure modes were usually not described adequately. Considering the limitation of the above studies, we aim to propose a new continuum damage model for 3D woven composites. In this model, the fiber fracture, inter-fiber fracture and matrix fracture, respectively, can be considered adequately. The failure of a type of 3D woven composites is predicted by this damage model under tension. In order to verify the accuracy of the prediction from this damage model, some typical quasi-static tensile experiments are carried out.

## 2. Continuum damage model

The material components of 3D woven composites include fiber and matrix. The structure consists of fiber yarns and matrix. The fiber yarn is formed by the fibers and relatively small amounts of permeated matrix. The fiber yarn is assumed to be transversely isotropic. In this section, the constitutive Continuum damage model is proposed for the fiber yarn and matrix of the 3D woven composites. There are three types of failure modes in the 3D woven composites: fiber failure, inter-fiber failure and matrix failure (Fig. 1). It is noted that the failure often occurs in the interface between the fiber yarn and matrix, but there is not interface material and it is the matrix failure, accurately.

### 2.1. Damage variables and constitutive functions

When the damage initiates and evolves in the material, the mechanical properties are degraded. The damage variables can be used to characterize the damage process. The constitutive function for the fiber yarns can be expressed as:

$$\boldsymbol{\varepsilon}_f = \mathbf{S}_f(d)\boldsymbol{\sigma}_f \tag{1}$$

where $\mathbf{S}_f(d)$ is the compliance matrix, which can be written as:

$$\mathbf{S}_f(d) = \begin{bmatrix} \frac{1}{(1-d_{f,1})E_{f,1}} & -\frac{\nu_{f,12}}{E_{f,1}} & -\frac{\nu_{f,13}}{E_{f,1}} & & & \\ & \frac{1}{(1-d_{f,2})E_{f,2}} & -\frac{\nu_{f,23}}{E_{f,2}} & & 0 & \\ & & \frac{1}{(1-d_{f,3})E_{f,3}} & & & \\ & & & \frac{1}{(1-d_{f,4})G_{f,12}} & & \\ & \text{sym.} & & & \frac{1}{(1-d_{f,5})G_{f,23}} & \\ & & & & & \frac{1}{(1-d_{f,6})G_{f,31}} \end{bmatrix} \tag{2}$$

where the subscript 1 denotes the longitudinal direction of the fiber yarn, the subscript 2 and 3 denote the transverse directions of the fiber yarn, $E_{f,1}, E_{f,2}, E_{f,3}, G_{f,12}, G_{f,23}, G_{f,31}, \nu_{f,12}, \nu_{f,13}$ and $\nu_{f,23}$ are the elastic properties of the fiber yarn, and $d_{f,i}(i = 1 \sim 6)$ is the damage variable. The damage variable $d_{f,1}$ is associated with fiber fracture, and $d_{f,2}$ and $d_{f,3}$ are associated with inter-fiber fracture in the fiber yarn. $d_{f,4}$ and $d_{f,6}$ are associated with fiber fracture and inter-fiber fracture, and $d_{f,5}$ is influenced by inter-fiber fracture. For the convenience of the following derivation, the effective stress $\tilde{\boldsymbol{\sigma}}_f$ can be written as:

$$\tilde{\boldsymbol{\sigma}}_f = \mathbf{S}_{f0}^{-1}(d)\boldsymbol{\varepsilon}_f \tag{3}$$

where $\mathbf{S}_{f0}(d)$ is the undamaged compliance matrix obtained from eq. (2) using $d_{f,i} = 0(i = 1 \sim 6)$.

Similarly, the constitutive function for the matrix can be expressed as:

$$\boldsymbol{\varepsilon}_{\boldsymbol{m}} = \mathbf{S}_{\boldsymbol{m}}(d)\boldsymbol{\sigma}_{\boldsymbol{m}} \tag{4}$$

where $\mathbf{S}_{\boldsymbol{m}}(d)$ is the compliance matrix, which can be written as:

$$\mathbf{S}_{\boldsymbol{m}}(d) = \frac{1}{E_m}\begin{bmatrix} \frac{1}{1-d_m} & -\nu_m & -\nu_m & & & \\ & \frac{1}{1-d_m} & -\nu_m & & 0 & \\ & & \frac{1}{1-d_m} & & & \\ & & & \frac{2(1-\nu_m)}{1-d_m} & & \\ & \text{sym.} & & & \frac{2(1-\nu_m)}{1-d_m} & \\ & & & & & \frac{2(1-\nu_m)}{1-d_m} \end{bmatrix} \tag{5}$$

where $E_m, G_m$ and $\nu_m$ are the elastic properties the matrix, and $d_m$ is the damage variable which is associated with matrix fracture. The effective stress $\tilde{\boldsymbol{\sigma}}_m$ can be written as:

$$\tilde{\boldsymbol{\sigma}}_m = \mathbf{S}_{m0}^{-1}(d)\boldsymbol{\varepsilon}_m \tag{6}$$

where $\mathbf{S}_{m0}(d)$ is the undamaged compliance matrix obtained from eq. (5) using $d_m = 0$.

In order to differentiate the effects of the stress state on the failure modes, the damage variables are introduced as follows:

$$d_{f,1} = \begin{cases} d_{f,1t} \text{ if } \tilde{\sigma}_{f,11} \geqslant 0 \\ d_{f,1c} \text{ if } \tilde{\sigma}_{f,11} < 0 \end{cases}, \quad d_{f,i} = \begin{cases} d_{f,it} \text{ if } \tilde{\sigma}_{f,n} \geqslant 0 \\ d_{f,ic} \text{ if } \tilde{\sigma}_{f,n} < 0 \end{cases} (\mathrm{i} = 2, 3),$$
$$d_m = \begin{cases} d_{m,t} \text{ if } I_1 \geqslant 0 \\ d_{m,c} \text{ if } I_1 < 0 \end{cases} \tag{7}$$

where $\tilde{\sigma}_{f,11}$ is the effective stress in the longitudinal direction of the fiber yarn. The effective stress $\tilde{\sigma}_{f,n}$ and the invariant $I_1$ of stress tensor will be expressed in the following paragraphs.

### 2.2. Damage initiation and propagation criteria

To predict the damage initiation and propagation in the fiber yarn and matrix and evaluate the mechanical properties, the damage initiation and propagation criteria can be defined as follows:

$$F_{f,N} = \phi_{f,N} - r_{f,N} \leqslant 0, \quad N = \{1t, 1c, 2t, 2c, 3t, 3c\} \tag{8a}$$

$$F_{m,L} = \phi_{m,L} - r_{m,L} \leqslant 0, \quad L = \{t, c\} \tag{8b}$$

where $\phi_{f,N}$ is the loading function for different failure modes for the fiber yarn, $\phi_{m,L}$ for the matrix, and $r_{f,N}$ and $r_{m,L}$ are the damage threshold parameters which are dependent on the damage level and the loading history. The initial values of $r_{f,N}$ and $r_{m,L}$ are 1 and increase when the damage initiates and accumulates. The Puck failure criteria [7,8] is adopted for the fiber yarn as it can predict not only the location but also the orientation of a crack. The loading functions for the fiber yarn can be expressed as:

$$\phi_{f,1t} = \frac{\varepsilon_{f,11}E_{f1} + m_f\nu_{f,12}\tilde{\sigma}_{f,22} + m_f\nu_{f,13}\tilde{\sigma}_{f,33}}{S_{f,1t}} \text{ for } \tilde{\sigma}_{f,11} \geqslant 0 \tag{9a}$$

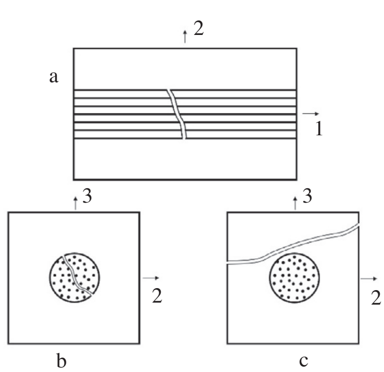


**Fig. 1.** The failure modes of the 3D woven composites: (a) fiber fracture, (b) inter-fiber fracture and (c) matrix fracture. Axis 1 is along the longitudinal direction of the fiber yarn, axis 2 and 3 are the transverse directions.

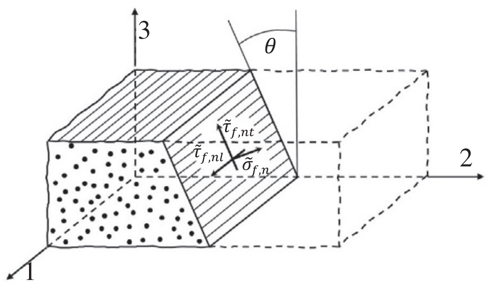


**Fig. 2.** The action plane in the fiber yarn. Axis 1 is along the longitudinal direction of the fiber yarn, axis 2 and 3 are the transverse directions.

$$\phi_{f,1c} = -\frac{\varepsilon_{f,11}E_{f1} + m_f\nu_{f,12}\tilde{\sigma}_{f,22} + m_f\nu_{f,13}\tilde{\sigma}_{f,33}}{S_{f,1c}} \quad \text{for } \tilde{\sigma}_{f,11} < 0 \tag{9b}$$

$$\begin{cases} \phi_{f,2t} = 1 + \left[\phi_{f,t}^{max}(\theta') - 1\right]cos^2\theta' \\ \phi_{f,3t} = 1 + \left[\phi_{f,t}^{max}(\theta') - 1\right]sin^2\theta' \end{cases} \quad \text{for } \tilde{\sigma}_{f,n} \geqslant 0 \tag{9c}$$

$$\begin{cases} \phi_{f,2c} = 1 + \left[\phi_{f,t}^{max}(\theta') - 1\right]cos^2\theta' \\ \phi_{f,3c} = 1 + \left[\phi_{f,t}^{max}(\theta') - 1\right]sin^2\theta' \end{cases} \quad \text{for } \tilde{\sigma}_{f,n} < 0 \tag{9d}$$

where $m_f$ is the 'stress magnification effect' caused by the different moduli of fibers and matrix, $S_{f,1t}$ and $S_{f,1c}$ is the tensile and compressive strengths of the fiber along the fiber direction, the angle $\theta'$ is used to denote the most dangerous plane, and the loading functions $\phi_{f,t}^{max}(\theta')$ and $\phi_{f,c}^{max}(\theta')$ can be written as:

$$\begin{cases} \phi_{f,t}^{max}(\theta') = \max\limits_{\theta\in[0,\pi)}\{\phi_{f,t}(\theta)\} \\ \phi_{f,t}(\theta) = \sqrt{\left[\tilde{\sigma}_{f,n}\left(\frac{1}{S_{f,2t}^A} - \frac{p_{\phi,t}}{S_{f,\phi}^A}\right)\right]^2 + \left(\frac{\tilde{\tau}_{f,nt}}{S_{f,23}^A}\right)^2 + \left(\frac{\tilde{\tau}_{f,nl}}{S_{f,21}^A}\right)^2} + \tilde{\sigma}_{f,n}\frac{p_{\phi,t}}{S_{f,\phi}^A} \end{cases} \tag{10a}$$

$$\begin{cases} \phi_{f,c}^{max}(\theta') = \max\limits_{\theta\in[0,\pi)}\{\phi_{f,c}(\theta)\} \\ \phi_{f,c}(\theta) = \sqrt{\left(\frac{\tilde{\tau}_{f,nt}}{S_{f,23}^A}\right)^2 + \left(\frac{\tilde{\tau}_{f,nl}}{S_{f,21}^A}\right)^2 + \left(\tilde{\sigma}_{f,n}\frac{p_{\phi,c}}{S_{f,\phi}}\right)^2} + \tilde{\sigma}_{f,n}\frac{p_{\phi,c}}{S_{f,\phi}} \end{cases} \tag{10b}$$

where the angle $\theta$ is used to denote the action plane (Fig. 2), $\theta = \theta'$ when $\phi_{f,t}(\theta)$ or $\phi_{f,c}(\theta)$ reaches its maximum value, $S_{f,2t}^A, S_{f,21}^A, S_{f,23}^A$ and $S_{f,\phi}^A$ are fracture resistance. Here, $S_{f,2t}^A = S_{f,2t}, S_{f,21}^A = S_{f,21}$ and $S_{f,23}^A = \frac{S_{f,2c}}{2(1+p_{23,c})}$. $S_{f,2t}$ and $S_{f,2c}$ are the tensile and compressive strengths of the fiber yarns transverse to the fiber direction. $S_{f,21}$ is the shear strength of the fiber yarn transverse and parallel to the fiber direction. $p_{\phi,t}, p_{\phi,c}, p_{23,c}$ and $p_{23,t}$ are the inclination parameters. $\tilde{\sigma}_{f,n}, \tilde{\tau}_{f,nt}$ and $\tilde{\tau}_{f,nl}$ are the normal and shear stresses on the action plane and can be expressed as:

$$\begin{cases} \tilde{\sigma}_{f,n} = \tilde{\sigma}_{f,22}cos^2\theta + \tilde{\sigma}_{f,33}sin^2\theta + 2\tilde{\tau}_{f,23}sin\theta cos\theta \\ \tilde{\tau}_{f,nt} = (\tilde{\sigma}_{f,33} - \tilde{\sigma}_{f,22})sin\theta cos\theta + \tilde{\tau}_{f,23}(cos^2\theta - sin^2\theta) \\ \tilde{\tau}_{f,nl} = \tilde{\tau}_{f,31}sin\theta + \tilde{\tau}_{f,21}cos\theta \end{cases} \tag{11}$$

For the matrix, the paraboloidal yield criterion [22] is adopted, thus the loading functions for the matrix can be written as:

$$\begin{cases} \phi_{m,t} = \frac{3J_2 + I_1\left(S_{m,c} - S_{m,t}\right)}{S_{m,c}S_{m,t}} & \text{for } I_1 \geqslant 0 \\ \phi_{m,c} = \frac{3J_2 + I_1\left(S_{m,c} - S_{m,t}\right)}{S_{m,c}S_{m,t}} & \text{for } I_1 < 0 \end{cases} \tag{12}$$

where $S_{m,t}$ and $S_{m,c}$ are the tensile and compressive yield strengths of the matrix, $I_1$ is the first invariant of the corresponding stress tensor, and $J_2$ are the second invariant of the corresponding deviatoric stress tensor. $I_1$ and $J_2$ can be written as:

$$\begin{cases} I_1 = \tilde{\sigma}_{m,11} + \tilde{\sigma}_{m,22} + \tilde{\sigma}_{m,33} \\ J_2 = \frac{1}{6}\left[(\tilde{\sigma}_{m,11} - \tilde{\sigma}_{m,22})^2 + (\tilde{\sigma}_{m,22} - \tilde{\sigma}_{m,33})^2 + (\tilde{\sigma}_{m,33} - \tilde{\sigma}_{m,11})^2\right] \end{cases} \tag{13}$$

### 2.3. Damage evolution law

In the damage initiation and propagation criteria, when $F(F_{f,N}$ or $F_{m,L})$ is less than 0, the material is undamaged. When $F$ is equal to 0, the material is damaged. The damage threshold value $r(r_{f,N}$ or $r_{m,L})$ must satisfy the Kuhn–Tucker and consistency conditions:

$$\dot{r} \geqslant 0; \quad F \leqslant 0; \quad \dot{r}F = 0 \tag{14}$$

$$\dot{F} = 0, \quad \text{if } F = 0 \tag{15}$$

Based on the above conditions and Eq. (8), we obtain:

$$\begin{cases} \dot{r} = 0, & \text{if } F < 0 \\ \dot{r} = \dot{\phi}, & \text{if } F = 0 \end{cases} \tag{16}$$

Integrating Eq. (16), the damage threshold value $r_{f,N}$ for the fiber yarn can be expressed as:

$$r_{f,N} = \max\left\{1, \max\left\{\phi_{f,N}^{\tau}\right\}\right\}, \quad N = \{1t, 1c, 2t, 2c, 3t, 3c\}, \tau \in [0, t] \tag{17}$$

where $t$ is the total time. In the same way, we can obtain:

$$r_{m,L} = \max\left\{1, \max\left\{\phi_{m,L}^{\tau}\right\}\right\}, \quad L = \{t, c\}, \tau \in [0, t] \tag{18}$$

The damage process is irreversible and the mechanical properties are degraded as soon as any one criterion is activated. The exponential damage evolution laws [23] (Fig. 3(a)) are adopted for the fiber yarn, except for the longitudinal tension failure modes. The linear and exponential damage evolution laws (Fig. 3(b)) are used to represent the phenomena of fiber bridging and fiber pull-out in the fiber yarn [14,24]. The damage variables of the fiber yarn can be expressed as:

$$d_{f,1t} = 1 - \frac{1 - d_{f,1t}^l}{r_{f,1t}^e}\exp\left[A_{f,1t}\left(1 - r_{f,1t}^e\right)\right] \tag{19a}$$

$$d_{f,N} = 1 - \frac{1}{r_{f,N}}\exp\left[A_{f,N}(1 - r_{f,N})\right], \quad N = \{1c, 2t, 2c, 3t, 3c\} \tag{19b}$$

where $A_{f,N}$ is a parameter that defines the exponential softening law. $r_{f,1t}^e, A_{f,1t}$ and $d_{f,1t}^l$ are the auxiliary damage threshold parameter, softening law parameter and damage variable. They can be expressed as:

$$\begin{cases} d_{f,1t}^l = \left(1 + \frac{K_{f,1}}{E_{f,1}}\right)\left(1 - \frac{1}{r_{f,1t}^l}\right) \\ r_{f,1t}^l = \max\left[1, \min\left(r_{f,1t}, r_{f,1t}^F\right)\right] \\ r_{f,1t}^e = \max\left[1, \left(1 - r_{f,1t}^f\right)\frac{S_{f,1t}}{S_{po}}\right] \end{cases} \tag{20}$$

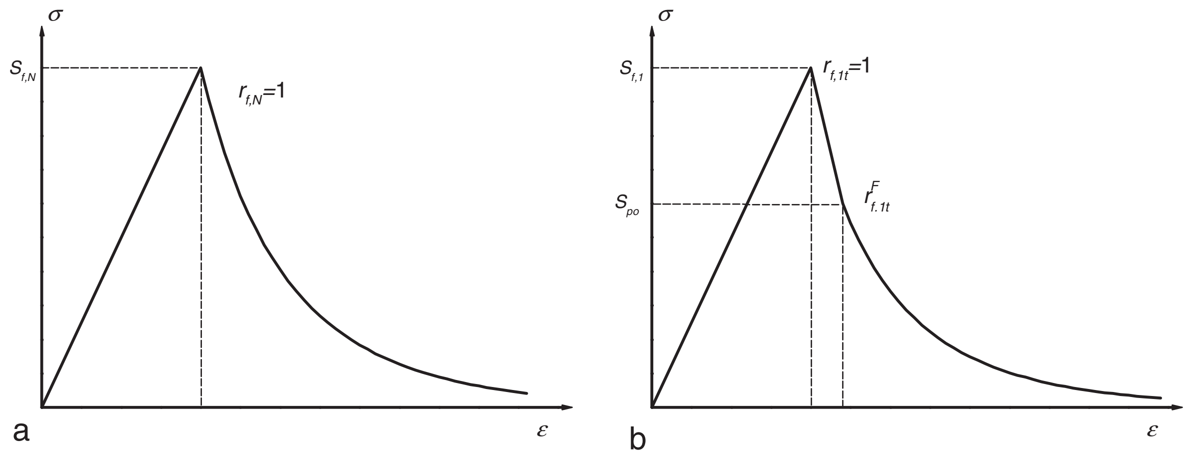


**Fig. 3.** The damage evolution of the fiber yarn: (a) the exponential damage evolution law and (b) the linear and exponential damage evolution law.

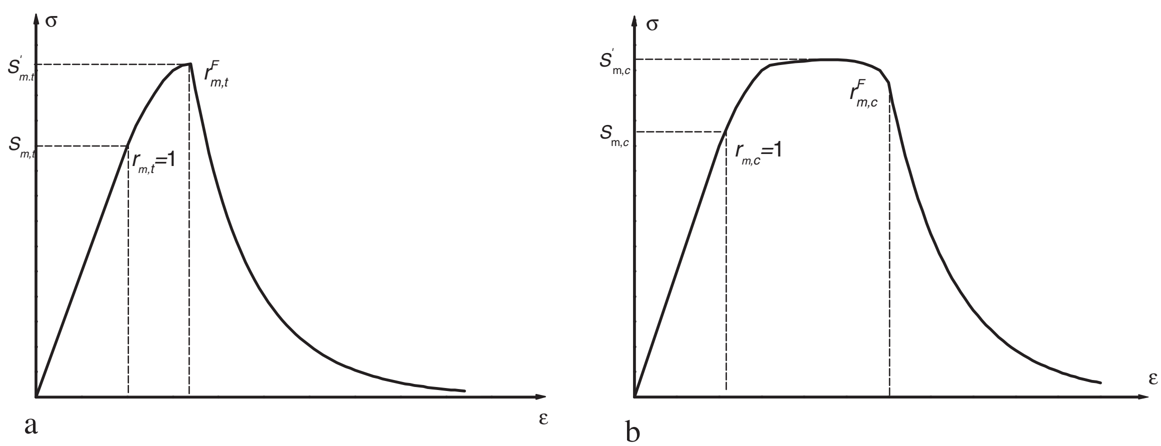


**Fig. 4.** Damage evolution of the matrix under tension (a) and compression (b).

where $K_{f,1}$ is the slope of linear softening law, the auxiliary damage threshold value $r^F_{f,1t}$ and the limit stress $S_{po}$ are on the intersection of linear and exponential laws (Fig. 3(b)).

As mentioned previously, the damage variables $d_{f,4}$ and $d_{f,6}$ represent the damage effects on the shear stiffness due to fiber and inter-fiber fracture, and $d_{f,5}$ due to inter-fiber fracture. Under these circumstances, the damage variables $d_{f,4}, d_{f,5}$ and $d_{f,6}$ can be given by

$$\begin{aligned} d_{f,4} &= 1-(1-d_{f,1})(1-d_{f,2}) \\ d_{f,5} &= 1-(1-d_{f,2})(1-d_{f,3}) \\ d_{f,6} &= 1-(1-d_{f,3})(1-d_{f,1}) \end{aligned} \tag{21}$$

The failure modes of the matrix are different from those of the fiber yarn. According to the experimental characteristics of reference [20], the exponential damage evolution laws (Fig. 4) are developed and modified for the matrix as follow:

$$d_{m,L} = 1 - \frac{1-d^l_{m,L}}{r^e_m}\exp\left[A_{m,L}(1-r^e_m)\right], \quad L=\{t,c\} \tag{22}$$

where $r^e_m, A_{m,L}$ and $d^l_{m,L}$ are the auxiliary damage threshold parameter, softening law parameter and damage variable. They can be expressed as:

$$\begin{cases} d^l_{m,L} = A^*_{m,L}\left[\sqrt{\exp\left(r^l_m-1\right)}-1\right] \\ r^l_m = \max\left[1, \min\left(r_m, r^F_{m,L}\right)\right] \\ r^e_m = \max\left[1, \frac{r_m}{r^F_{m,L}}\right] \end{cases} \tag{23}$$

where $r^l_m$ and $A^*_{m,L}$ are the auxiliary damage threshold parameter and softening law parameter, $r^F_{m,L}$ is the damage threshold parameter at the transition between the exponential and auxiliary damage evolution laws (Fig. 4).

In order to ensure that the computed dissipated energy of all the microcracks per unit area of the fracture surface is independent of mesh size of the finite elements, all softening law parameter $A(A_{f,N}, A_{m,L}$ and $A^*_{m,L})$ is obtained by regularizing the softening branch of the stress–strain curve. Based on the Bazant's crack band theory [25], the energy dissipated per unit volume $g_M$ can be determined by the fracture toughness $G_M$:

$$g_M = \frac{G_M}{l^*}, \quad M \in \{f,1t; f,1c; f,2t; f,2c; f,3t; f,3c; m,t; m,c\} \tag{24}$$

where $l^*$ is the characteristic length of the finite element. The energy dissipated per unit volume can be obtained from the integration of the damage energy dissipation during the process of damage development:

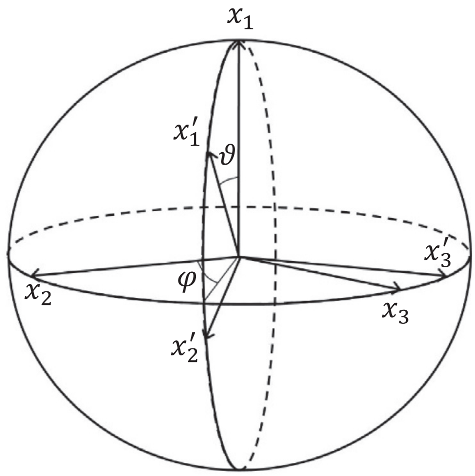


**Fig. 5.** Definition of the Euler angles.

$$g_M = \int_0^\infty \frac{\partial G}{\partial d_M}\dot{d}_M dt = \int_1^\infty \frac{\partial G}{\partial d_M}\frac{\partial d_M}{\partial r_M} dr_M \tag{25}$$

where the Helmholtz free energy $G$ can be written as:

$$G = \frac{1}{2}\boldsymbol{\sigma\varepsilon} \tag{26}$$

According to Eq. (26), the Helmholtz free energy for the fiber yarn and matrix can be written as:

$$\begin{aligned} G_f = {} & \frac{\sigma_{f,11}^2}{2(1-d_{f,1})E_{f1}} + \frac{\sigma_{f,22}^2}{2(1-d_{f,2})E_{f2}} + \frac{\sigma_{f,33}^2}{2(1-d_{f,3})E_2} \\ & - \frac{\upsilon_{f12}}{E_{f1}}\sigma_{f,11}\sigma_{f,22} - \frac{\upsilon_{f13}}{E_{f1}}\sigma_{f,11}\sigma_{f,33} - \frac{\upsilon_{f23}}{E_{f2}}\sigma_{f,22}\sigma_{f,33} \\ & + \frac{\tau_{f,12}^2}{2(1-d_{f,4})G_{f12}} + \frac{\tau_{f,23}^2}{2(1-d_{f,5})G_{f23}} + \frac{\tau_{f,31}^2}{2(1-d_{f,6})G_{f,31}} \end{aligned} \tag{27}$$

$$\begin{aligned} G_m = {} & \frac{\sigma_{m,11}^2 + \sigma_{m,22}^2 + \sigma_{m,33}^2}{2(1-d_m)E_m} \\ & - \frac{\upsilon_m}{E_m}(\sigma_{m,11}\sigma_{m,22} + \sigma_{m,22}\sigma_{m,33} + \sigma_{m,33}\sigma_{m,11}) \\ & + \frac{\tau_{m,12}^2 + \tau_{m,23}^2 + \tau_{m,31}^2}{2(1-d_m)G_m} \end{aligned} \tag{28}$$

All the softening law parameter $A$ are determined by Eqs. (24)–(26), (27).

## 3. Finite element implementation

Each softening law parameter in the damage evolution functions needs to be calculated by the corresponding fracture toughness. $G_{f,1t}$ corresponds to the fracture toughness for the longitudinal model-I crack of the fiber yarn as well as $G_{f,2t}$ and

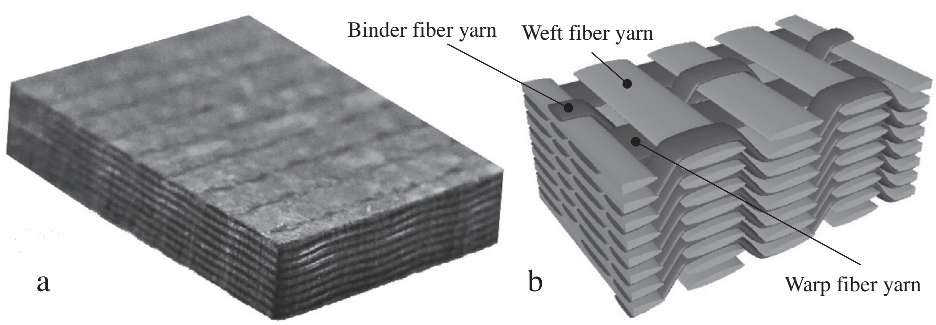


**Fig. 6.** The appearance (a) and geometry structure (b) of the 3D woven composite.

**Table 1**
The properties of the material constituents.

| | $E_{f,1}$ | $E_{f,2}$ | $G_{f,12}$ | $\nu_{f,12}$ | $\nu_{f,23}$ | $S_{f,1t}$ | Density |
|---|---|---|---|---|---|---|---|
| T700 | 230 GPa | 18.2 GPa | 36.6 GPa | 0.27 | 0.3 | 4.9 GPa | 1.80 g/cm$^3$ |
| T300 | 221 GPa | 13.8 GPa | 9.0 GPa | 0.20 | 0.25 | 3.53 GPa | 1.75 g/cm$^3$ |
| | $E_m$ | | | $\nu_m$ | | $S_{m,t}$ | Density |
| TDE-86 | 3.55 GPa | | | 0.33 | | 241 GPa | 1.22 g/cm$^3$ |

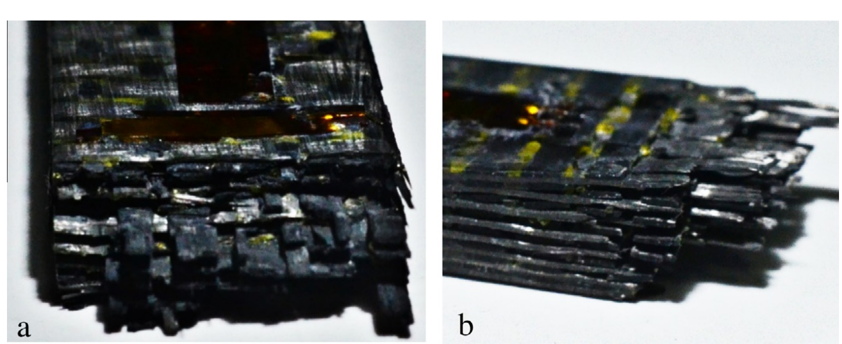


**Fig. 7.** Typical tensile failure modes of the specimens.

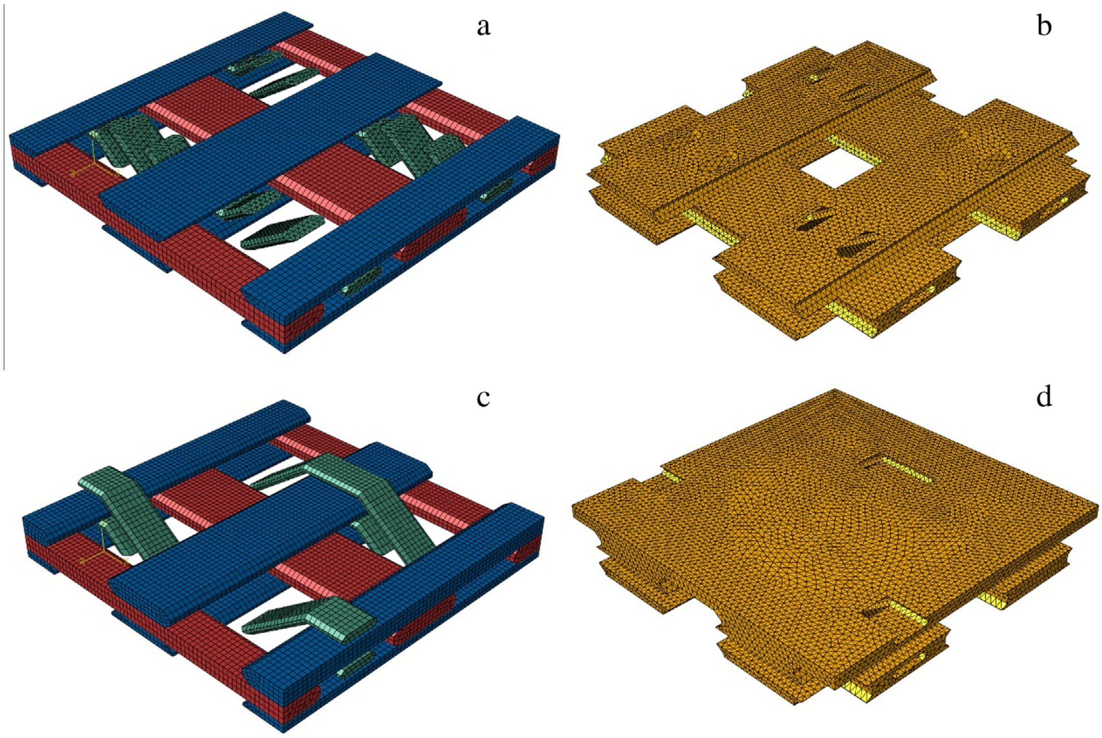


**Fig. 8.** The finite element model of the interior and surface cell of RVE in the 3D woven composite: the fiber yarns (a) and matrix (b) of the interior cell and the fiber yarns (c) and matrix (d) of the surface cell.

**Table 2**
Parameters of the 3D woven composites' structure.

| | Fiber | Fiber yarns | | | Sub unit cells | |
|---|---|---|---|---|---|---|
| | | Warp fiber yarn | Weft fiber yarn | Binder fiber yarn | Interior cell | Surface cell |
| Volume fraction (%) | 50.9 | 24.6 | 27.3 | 6.7 | 70.6 | 29.4 |

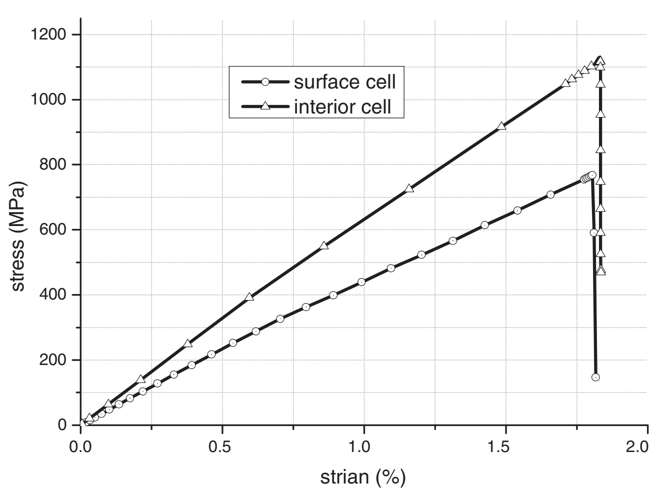


**Fig. 9.** The stress–strain curves of the interior and surface cells.

$G_{f,3t}$ correspond to the fracture toughness for the transverse model-I crack and the crack angle [14]. $G_{m,t}$ corresponds to the fracture toughness of the model-I crack in the matrix. $G_{f,1c}$ can be obtained

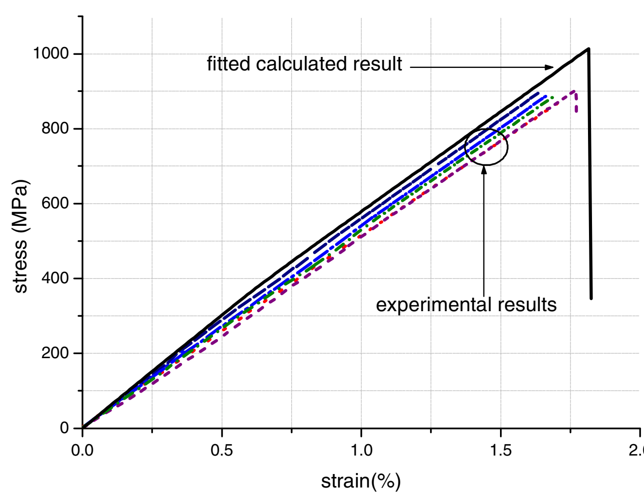


**Fig. 10.** The stress–strain curves of the fitted and experimental results.

from evaluation of the energy dissipated per unit area in a kind band that results from damage of supporting matrix [26]. $G_{f,2c}$ and $G_{f,3c}$ can be calculated approximately using the fracture toughness of the transverse crack in mode-II of the fiber yarn, the crack angle and a term accounting for friction between the crack faces [14,27]. $G_{m,c}$ can be obtained using the fracture toughness of the crack in mode-II of the matrix. Besides all the fracture toughness, the characteristic length $l^*$ is important for the softening law parameter. For hexahedron element, we take $l^* \approx \sqrt[3]{V_I}$, and for tetrahedron element, $l^* \approx \sqrt[3]{6\sqrt{2}V_I}$. $V_I$ is the volume associated with each integration point.

As mentioned in section 2, the fiber yarn is assumed to be transverse isotropic. The constitutive function is stated in the fiber

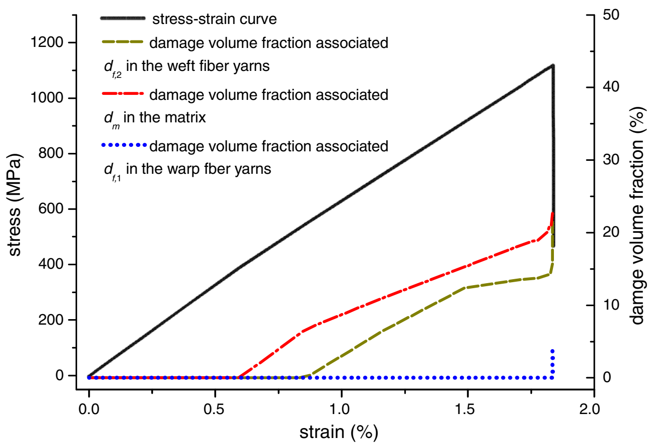


**Fig. 11.** Damage accumulation in the interior cell: the volume fraction of the damaged elements in the warp fiber yarns ($d_{f,1}$), weft fiber yarns ($d_{f,2}$) and matrix ($d_m$).

yarn's local coordinate system. So it is necessary to convert the stress and strain in global coordinate system to that in the local in the FE model.

$$\boldsymbol{\varepsilon}_f = \mathbf{T}\boldsymbol{\varepsilon}, \quad \boldsymbol{\sigma}_f = \mathbf{T}\boldsymbol{\sigma} \tag{29}$$

where $\mathbf{T}$ is the coordinate transform matrix and it can be expressed as

$$\mathbf{T} = \begin{bmatrix} Q_{11}^2 & Q_{21}^2 & Q_{31}^2 & Q_{11}Q_{21} & Q_{21}Q_{31} & Q_{31}Q_{11} \\ Q_{12}^2 & Q_{22}^2 & Q_{32}^2 & Q_{12}Q_{22} & Q_{22}Q_{32} & Q_{32}Q_{12} \\ Q_{13}^2 & Q_{23}^2 & Q_{33}^2 & Q_{13}Q_{23} & Q_{23}Q_{33} & Q_{33}Q_{13} \\ Q_{11}Q_{12} & Q_{21}Q_{22} & Q_{31}Q_{32} & Q_{11}Q_{22} & Q_{21}Q_{32} & Q_{31}Q_{12} \\ Q_{12}Q_{13} & Q_{22}Q_{23} & Q_{32}Q_{33} & Q_{12}Q_{23} & Q_{22}Q_{33} & Q_{32}Q_{13} \\ Q_{13}Q_{11} & Q_{23}Q_{21} & Q_{33}Q_{31} & Q_{13}Q_{21} & Q_{23}Q_{31} & Q_{33}Q_{11} \end{bmatrix} \tag{30}$$

where $Q_{ij}(i = 1,2,3, j = 1,2,3)$ is taken from the following matrix $\mathbf{Q}$ which can be written as

$$\mathbf{Q} = \begin{bmatrix} cos\vartheta & sin\vartheta cos\varphi & sin\vartheta sin\varphi \\ -sin\vartheta & cos\vartheta cos\varphi & cos\vartheta sin\varphi \\ 0 & -sin\phi & cos\phi \end{bmatrix} \tag{31}$$

where the angles $\vartheta$ and $\varphi$ called Euler angles. They define the relationship of the local and global coordinate systems (Fig. 5).

The proposed continuum damage model is implemented in finite element software Abaqus/Explicit using the user-defined subroutine VUMAT.

## 4. Results and discussion

In this section, the failure a type of 3D woven composites is predicted under quasi-static tension, and the corresponding experiments are carried out. The numerical and experimental results are compared to validate the continuum damage model.

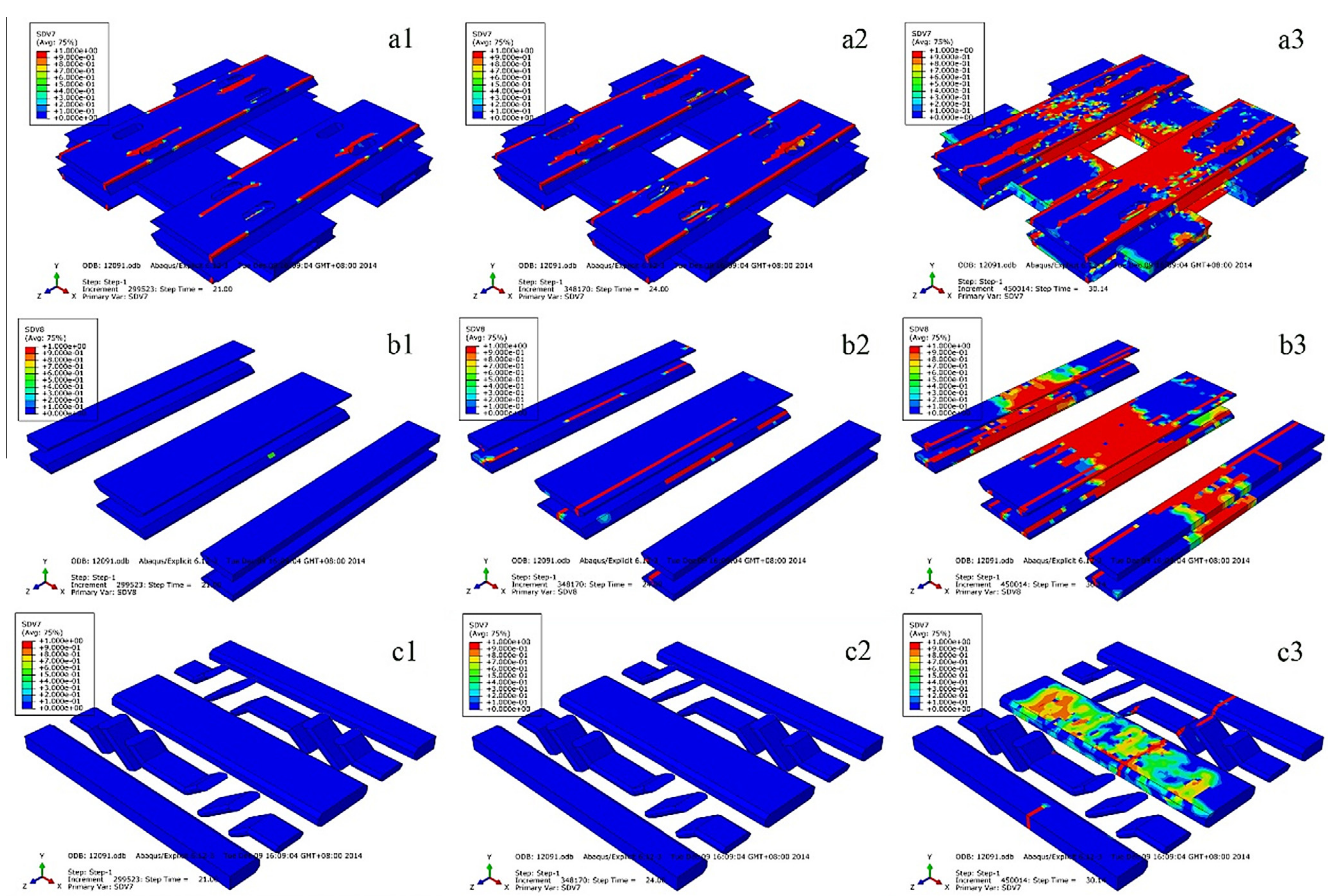


**Fig. 12.** Damage accumulation in the interior cell: damage associated with $d_m$ in the matrix (a1)–(a3), $d_{f,2}$ in the weft yarns (b1)–(b3) and $d_{f,1}$ in the warp and binder yarns. The average strain $\bar{\varepsilon}_x$ is equal to $8.7 \times 10^{-3}$, $1.14 \times 10^{-2}$ and $1.83 \times 10^{-2}$ for (a1, b1 and c1), (a2, b2 and c2) and (a3, b3 and c3), respectively.

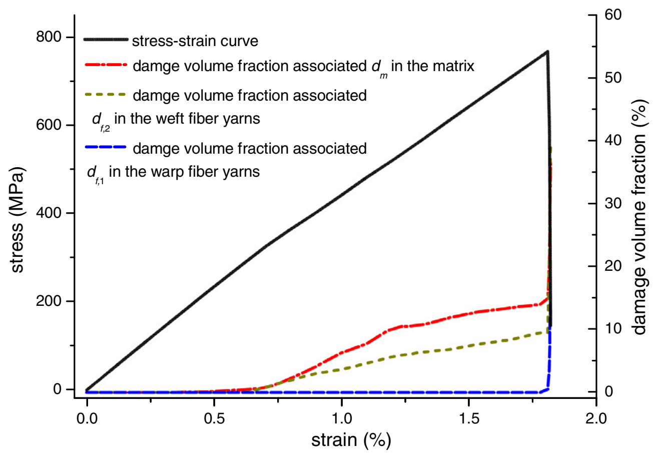


**Fig. 13.** Damage accumulation in the surface cell: the volume fraction of the damaged elements in the warp fiber yarns ($d_{f,1}$), weft fiber yarns ($d_{f,2}$) and matrix ($d_m$).

### 4.1. Quasi-static tensile experiments

The picture of the 3D woven composites and the structure of fiber yarns are shown in Fig. 6. The material constituents comprise 12 K T700 carbon fibers in the warp and weft yarns, 3 K T300 carbon fibers in the binder yarns and TDE-86 resin in the matrix. Their properties are shown in Table 1. The quasi-static uniaxial tensile experiments are done by referencing the testing standard ASTM D3039 on the Zwick/Roell Z100 universal testing machine. The length of the specimens is 250 mm, the width is 18 mm, the net length is 110 mm, and the thickness is 5 mm. All the experiments are operated at room temperature with a tension rate of 1 mm/min.

Fig. 7 shows the front and lateral of the fracture of the 3D woven composites after failure. From the picture, we know that the fracture appearance is not smooth and some warp fiber yarns are pulled out. The resin matrix between the fiber yarns near to the fracture, falls off the specimen.

### 4.2. Numerical calculation

In order to reduce the amount of the FEM calculation, the representative volume element (RVE) method is adopted. The RVE of the 3D woven composites has two types of sub unit cells: interior cell and surface cell. The finite element models of the interior and surface cells are built and shown in Fig. 8. The periodic boundary conditions are imposed upon the finite element model of the unit cell to assure correctness of the calculation. The periodic boundary conditions are carried out by utilizing the coupling displacement constraints which are applied upon the corresponding nodes of the opposite surfaces of the unit cell. The parameters of the structure are shown in Table 2.

Under tension in the warp direction, the stress–strain relationship of the interior and surface cells in the warp direction are obtained (Fig. 9). Depend on the volume fraction of the interior and surface cells in the structure, the stress–strain curve of the 3D woven composites is fitted and given in Fig. 10 in comparison with the experimental results. The initial slope and peak value of the fitted simulated curve are all slightly higher than experimental data. So the predicted elastic modulus and failure strength correlate well with the experimental ones. Figs. 11–14 show the damage accumulation in the interior and surface cells. Under the tensile

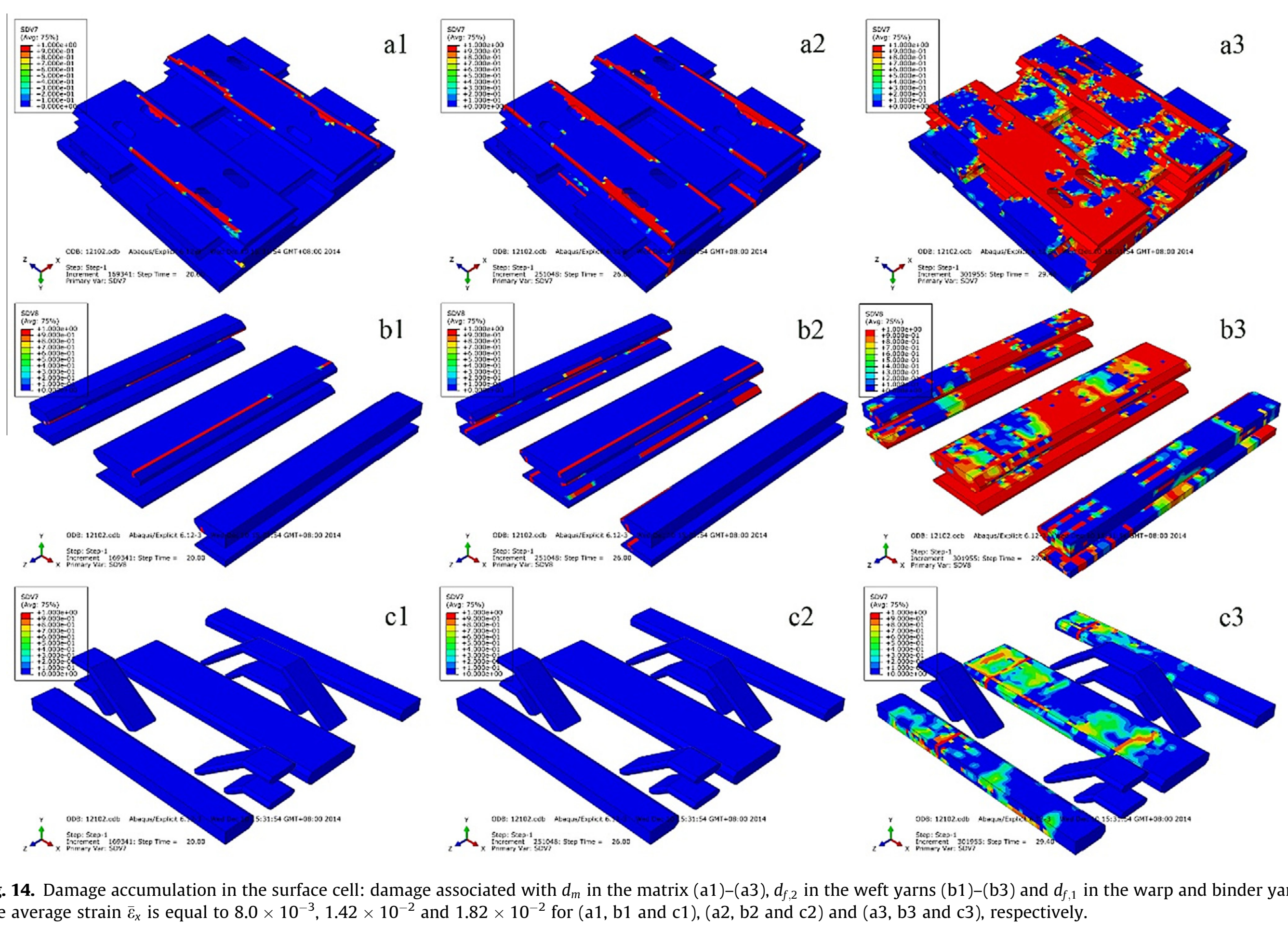


**Fig. 14.** Damage accumulation in the surface cell: damage associated with $d_m$ in the matrix (a1)–(a3), $d_{f,2}$ in the weft yarns (b1)–(b3) and $d_{f,1}$ in the warp and binder yarns. The average strain $\bar{\varepsilon}_x$ is equal to $8.0 \times 10^{-3}$, $1.42 \times 10^{-2}$ and $1.82 \times 10^{-2}$ for (a1, b1 and c1), (a2, b2 and c2) and (a3, b3 and c3), respectively.

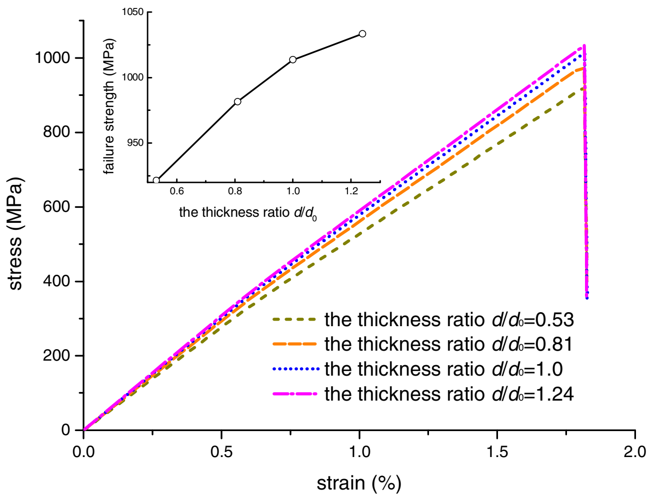


**Fig. 15.** Influence on stress–strain relationship by the thickness of the 3D woven composites.

load in the warp direction, the damage associated with the damage variables $d_{f,1}$ in the warp fiber yarns, $d_{f,2}$ in the weft fiber yarns and $d_m$ in the matrix, initiates and continuously accumulates. From the pictures, we know that the onset of the damage in the matrix is close to the fiber yarns. The damage in the weft yarns and matrix initiates at the strain (0.5%, 1%), and increases gradually. As a result, the slope of the stress–strain curve decreases a little. The damage in the warp fiber yarns initiates at the strain about 1.8% and increases abruptly and the stress–strain curve decrease suddenly. The damage in the warp fiber yarns causes failure of the cells and this phenomenon agrees well with the experimental results.

Because of the volume fraction of the warp fiber yarns in the interior cell is larger than that in the surface cell, so the slope of the stress–strain curve of the interior is larger than that in the surface cell too. It is found that the thickness of the 3D woven composites, associated with the volume ratio of the interior and surface cells, influences the mechanical properties of the 3D woven composites. The stress–strain curves of the 3D woven composites with typical thickness are calculated under tension in the warp direction and shown in Fig. 15. When the thickness increases, the slope of stress–strain curve and the failure strength raise.

## 5. Conclusion

A continuum damage model for the prediction of the damage initiation and development in the 3D woven composites is proposed. The onset of the fiber and inter-fiber failure in the fiber yarn are predicted using the Puck criteria and the onset of the matrix failure in the matrix are predicted using the paraboloidal yield criterion. A thought of damage evolution simulation is proposed for the 3D woven composites by combining different failure criteria and evolution laws in the level of the fiber yarn and the matrix. The advance of the present model is: the fiber failure, inter-fiber failure and matrix failure can be considered separately and adequately.

The damage model is implemented using the FEM. The prediction of the failure of a type of woven composites under tension agrees well with the corresponding experiments. Under tension, the damage in the fiber yarns parallel to loading direction occurs suddenly and causes the ultimate failure, and before the ultimate failure, the damage in the other fiber yarns and the matrix, initiates and develops gradually. Because of the fiber yarns volume fraction is different in the interior and surface cells and the volume ratio of interior and surface is related with the thickness, so the mechanical properties of the 3D woven composites are influenced by the thickness. The slope of stress–strain curve and the failure strength raise with the thickness increasing.

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